Result & Conclusion

Cases:

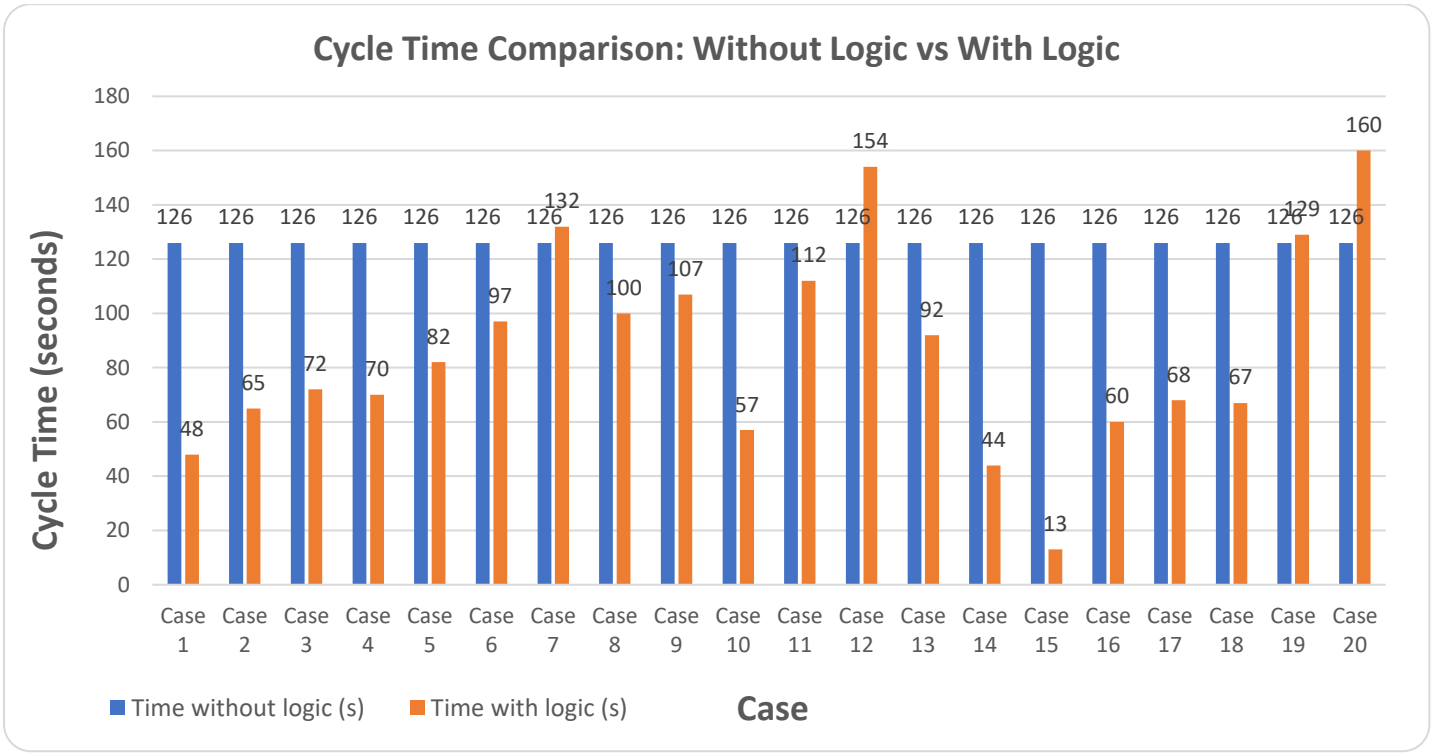
1. Less congestion, No E.V.
2. Less congestion, 1 E.V.
3. Less congestion, 2 E.V.
4. Less congestion, 2 E.V. in same lane
5. More congestion, No E.V.
6. More congestion, 1 E.V. Green Time **> 10 seconds for the previous lane; it will start from the lane next to the previous lane.**
7. More congestion, 1 E.V. Green Time **< 10 seconds for the previous lane.**
8. More congestion, 2 E.V.
9. More congestion, 2 E.V. in same lane
10. Bike count more, without E.V.
11. Bus count more, without E.V.
12. Truck count more, without E.V.
13. E.V. count more.
14. One lane is empty, and the other lanes are congested.
15. Only one lane has vehicles.
16. Less congestion, 1 E.V. Green Time **> 10 seconds for the previous lane; it will start from the lane next to the previous lane.**
17. Less congestion, 1 E.V. Green Time **< 10 seconds for the previous lane.**
18. Bike count more, with 1 E.V.
19. Bus count more, with 1 E.V.
20. Truck count more, with 1 E.V.

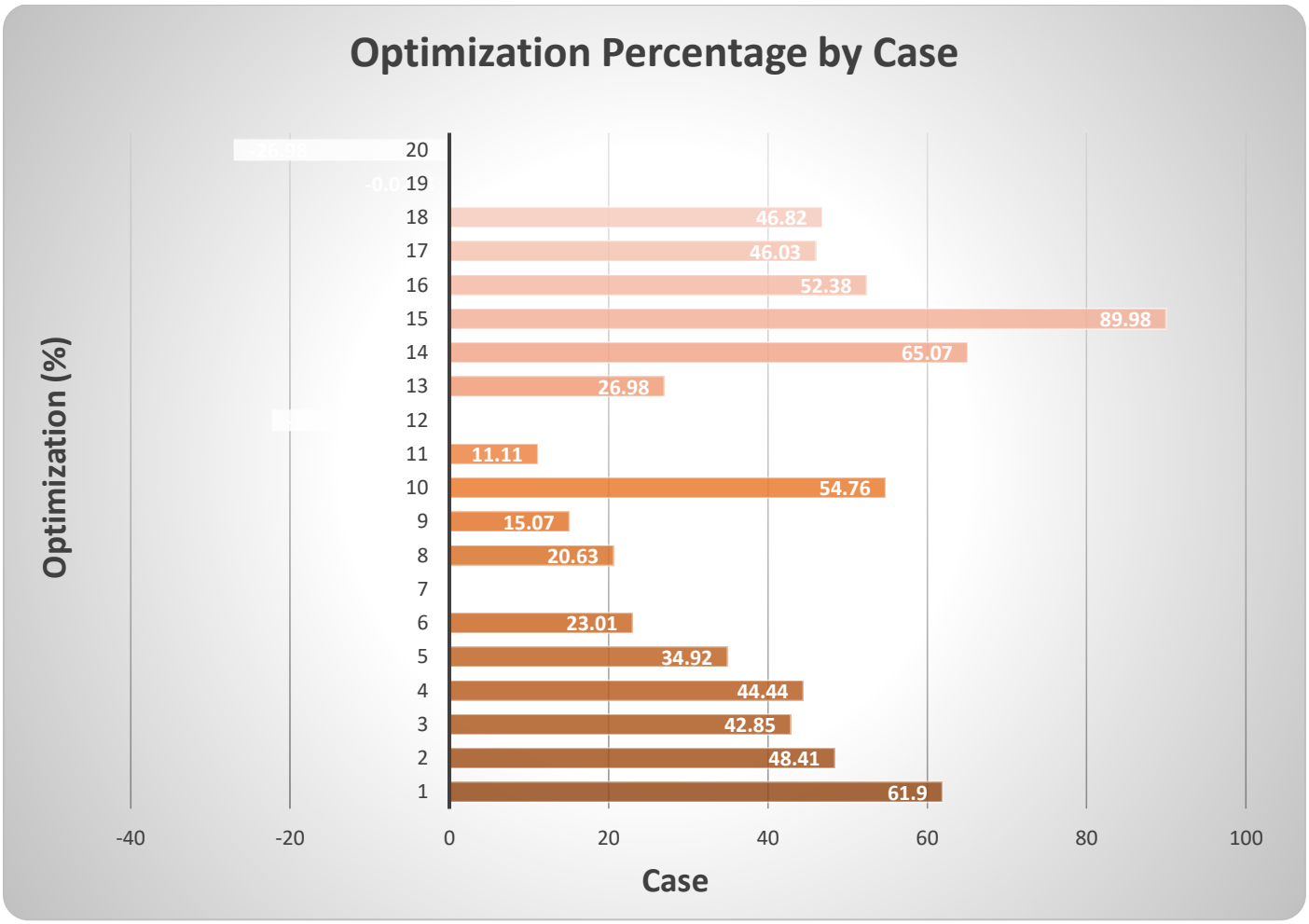
Observation Table :

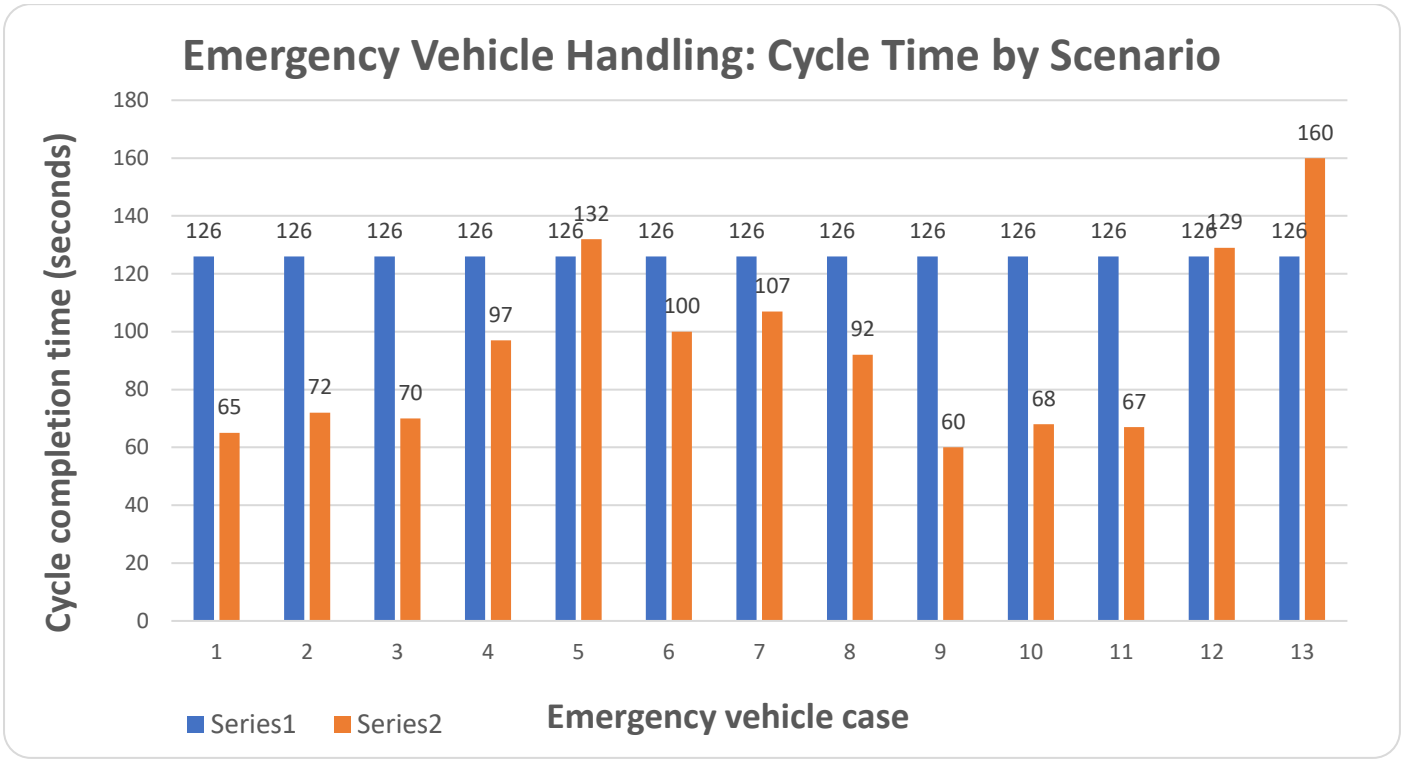
Result	Time without logic	Time with logic	optimization (%)
Case - 1	126	48	61.9
Case - 2	126	65	48.41
Case - 3	126	72	42.85
Case - 4	126	70	44.44
Case - 5	126	82	34.92
Case - 6	126	97	23.01
Case - 7	126	132	-0.0476
Case - 8	126	100	20.63
Case - 9	126	107	15.07
Case - 10	126	57	54.76
Case - 11	126	112	11.11
Case - 12	126	154	-22.22
Case - 13	126	92	26.98
Case - 14	126	44	65.07
Case - 15	126	13	89.98
Case - 16	126	60	52.38
Case - 17	126	68	46.03
Case - 18	126	67	46.82
Case - 19	126	129	-0.0238
Case - 20	126	160	-26.98
	Avg = 126	Avg = 86.45	Avg = 31.75 %

$$\textbf{Optimization (\%)} = \frac{\textbf{Time without logic} - \textbf{Time with logic}}{\textbf{Time without logic}} \times 100$$

Graphs :


Cycle Time Comparison: Without Logic vs With Logic
Cycle Time (seconds)
180
160
140
120
100
80
60
40
20
0
126 126 126 126 126 126 126 126 126 126 126 126 126 126 126 126 126 126 126 126
48 65 72 70 82 97 132 100 107 57 112 154 92 44 13 60 68 67 129 160
Case 1 Case 2 Case 3 Case 4 Case 5 Case 6 Case 7 Case 8 Case 9 Case 10 Case 11 Case 12 Case 13 Case 14 Case 15 Case 16 Case 17 Case 18 Case 19 Case 20
Time without logic (s)
Time with logic (s)
Case


Optimization Percentage by Case
Optimization (%)
20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1
46.82
46.03
52.38
89.98
65.07
26.98
11.11
54.76
15.07
20.63
23.01
34.92
44.44
42.85
48.41
61.9
-40 -20 0 20 40 60 80 100
Case


Emergency Vehicle Handling: Cycle Time by Scenario
Cycle completion time (seconds)
180
160
140
120
100
80
60
40
20
0
126 126 126 126 126 126 126 126 126 126 126 126 126
65 72 70 97 132 100 107 92 60 68 67 129 160
1 2 3 4 5 6 7 8 9 10 11 12 13
Series1
Series2
Emergency vehicle case

d